Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 2

Date: _____ Name: _____

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**35**
Part 2	**30**
CW	**5**
Marks Obtained	

Section A: Multiple Choice Questions (MCQ) (8x1=8 marks)

For each question 1-8, four options are given. Circle the correct option i, ii, iii or iv.

1. When a large file is sent across a network, it is split into packets. What is the typical size of each packet?

i. 1 Byte **ii.** 64 KiB
iii. 10 GiB **iv.** 500 MiB

2. In packet switching, which network device reads the destination IP address and directs each packet?

i. Monitor **ii.** Keyboard
iii. Router **iv.** Scanner

3. A standard USB cable consists of a four-wire shielded cable. What are the 4 wires used for?

i. 4 wires for power only **ii.** 2 wires for power and 2 wires for data
iii. 4 wires for sound only **iv.** 1 wire for power and 3 for video

4. What is the maximum recommended cable length for a standard USB connection before repeaters are needed?

i. 5 metres **ii.** 500 metres
iii. 5 kilometres **iv.** 100 kilometres

5. Which transmission method sends data bits one after another down a single channel or wire?

i. Parallel data transmission **ii.** Serial data transmission
iii. Skewed data transmission **iv.** Analogue transmission

6. Where is Parallel data transmission most commonly used because distances are very short?

i. Transatlantic sea cables **ii.** Internal buses on a motherboard
iii. Satellite links **iv.** Worldwide internet

7. Which value in a packet header stops a lost packet from bouncing around a network forever?

i. Hop number (TTL) **ii.** Colour depth
iii. Sample rate **iv.** Plug and play

8. Which part of a data packet contains the cyclic redundancy check (CRC) used for error checking?

i. Packet Payload **ii.** Packet Header
iii. Packet Trailer **iv.** Router Switch

Section B: Matching Terms (6x1=6 marks)

9. Match each term in Column A with its correct description in Column B:

Column A (Term)	Column B (Description)	Answer
i. Payload	A. Marks the end of the packet and contains error-checking data (CRC).	i → ____
ii. Packet Trailer	B. Bits arrive out of synchronization across parallel wires over long distances.	ii → ____
iii. Bit Skew	C. The actual user data being carried inside the packet (about 64 KiB).	iii → ____
iv. Crosstalk	D. Computer automatically detects a USB device and loads its driver.	iv → ____
v. Plug and Play	E. Electromagnetic interference between adjacent wires in parallel cables.	v → ____
vi. Packet Header	F. Contains sender IP address, destination IP address, and sequence number.	vi → ____

Section C: Direction of Data Transmission **(3x2=6 marks)**

10. Complete the table below by stating the direction of data flow and one real-life example:

Mode	How Data Flows (1 mark each)	Real-Life Example (1 mark)
i. Simplex		
ii. Half-Duplex		
iii. Full-Duplex		

Section D: Serial vs. Parallel Transmission **(5x1=5 marks)**

11. Tick (✓) whether each statement describes Serial or Parallel data transmission:

No.	Statement / Feature	Serial (✓)	Parallel (✓)
i.	Bits are sent one after another along a single wire or channel.		
ii.	Multiple bits (e.g., 8 bits / 1 byte) are sent simultaneously across several wires.		
iii.	Most reliable over long distances because there is zero risk of bit skew.		
iv.	Used inside a computer motherboard (internal buses) over very short distances.		
v.	Used by Universal Serial Bus (USB) cables to connect external devices.		

Section E: Short Questions (Packet Switching & USB) **(2+3=5 marks)**

12. Answer the following questions on Packet Headers (2m) and USB (3m):

a) State any TWO items of information stored inside a Packet Header: (2 marks)

i.

_____

ii.

_____

b) State TWO advantages and ONE limitation of the Universal Serial Bus (USB) interface: (3 marks)

i. Advantage 1: _____

ii. Advantage 2: _____

iii. Limitation: _____